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Composite telescopic sight and sight mount

Abstract

A mount to be permanently bonded to an optical sighting apparatus. The mount includes a mount body having a first end portion and a second end portion with a central axis extending therebetween and formed of a first composite material and configured to be permanently bonded to the optical sighting apparatus. The mount further includes a rail receiver portion for receiving a mounting rail, wherein the rail receiver portion further includes: at least one flat portion forming a plane having a major axis perpendicular to the central axis; a first angled portion located on each end of a minor axis of the flat portion, the first angled portion being capable of receiving a respective angled portion of a receiver portion of the mounting rail; at least one second angled portion intersecting with at least one of the first angled portions; and a clamping portion with a threaded mechanism to clamp the rail receiver portion to the mounting rail.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/938,444, filed on Jul. 24, 2020, which is a continuation of U.S. patent application Ser. No. 15/711,613, filed on Sep. 21, 2017, now U.S. Pat. No. 10,739,110, which claims the benefit of priority from U.S. Provisional Patent Application No. 62/561,001, filed on Sep. 20, 2017, and U.S. Provisional Patent Application No. 62/420,307, filed on Nov. 10, 2016. The contents of the prior applications are incorporated herein by reference in their entirety.

BACKGROUND

(1) Telescopic sight systems (also interchangeably referred to herein as scope sighting systems or scopes) typically are formed of aluminum, magnesium, or metal alloys that may be machined or cast into a cylindrically shaped scope body; the cylindrical body housing various optical and mechanical components. A telescopic sight is generally mounted to a firearm, or some other mounting point through use of a scope mount. Typical scope mounts include a scope base that may be mounted to a firearm or other object and a plurality of scope rings, which mount to the scope base. The scope rings may have a circular opening corresponding to the outer circumference of the scope body for clamping to the cylindrical scope body.

(2) Scopes generally offer fixed or variable magnification and may include adjustments for windage and elevation. Many scope sighting systems further include a reticle for assisting a user with optical measurement and/or aiming. A reticle commonly consists of a plurality of fine lines, which may be stationary, movable mechanically, or movably projected onto a display field, such as a diffraction grid. Attached U.S. Patent Application Publication 2012/0113507 A1 discloses one example of a reticle formed by light projected onto a diffraction grid, which is incorporated by reference herein. Alternatively, reticles may be electronically produced on a screen projecting a virtual image; the virtual image being may be formed digitally based on a processed signal received from an imaging sensor.

(3) Reticles are most typically located in a focal plane, at a point between the objective lens or

lenses and the ocular lens or lenses. An erector lens assembly may further be located between the objective lens or ocular lens, and the reticle may be located between the objective lens and the erector and/or between the erector and ocular lens, depending on if a magnification of the reticle is desired. Reticles may be formed of a wire, etched in a glass or transparent substrate, or may be projected onto or emitted from a display or display field within the scope. Reticles may be viewed in relation to a real image, which may be magnified or enhanced through optics and placement of the reticle in relation to the optics, or may be viewed in relation to a virtual image formed by a processor using light data received by an image sensor. A virtual image may be displayed using a backlit liquid crystal display (LCD) or transmission-type organic light-emitting diode (OLED) display. While the technology of scope sighting systems has improved over the years, a number of shortcomings are still present in the current mounting systems and reticle configurations.

(4) Common problems associated with typical mounting systems include: limitations on scope size and form factor due to a necessity for compatibility with standard ring sizes, corrosion between the scope rings and the scope body or between other mating surfaces in the mount assembly, misalignment of the mounts due to the thermal expansion qualities of different materials, user error in assembling the mounting system, and an increase in complexity and weight of the assembly.

(5) Common reticle systems include many shortcomings, the most common being: failure of the mechanical components required for movement of the reticle, misalignment due to shock or thermal expansion, and an increase in complexity and weight due to mechanical components housed inside the scope body. Further, in projected reticle systems, similar disadvantages exist; along with a further increase in complexity and potential for failure. In reticles that are electronically produced on a screen displaying a virtual image, several of the above disadvantages exist, with the addition of the risk that the failure of the electronic components may result in a black-out and inability to use the scope optics.

SUMMARY

(6) One aspect of the present disclosure relates to a composite or carbon fiber scope body having a mounting system integrally attached to the scope body. The integral mount may further include a clamping portion having a self-centering feature for reducing a lateral shift when mounting the scope to a mounting rail.

(7) Another aspect of the present disclosure relates to a sighting system that includes at least one transparent and segmented digital display capable of emitting light from the display. The digital display may display a reticle and/or other pertinent information. The transparent and segmented electroluminescent display may include a thin film substrate produced on an optical or other transparent component within the scope body. The display or plurality of displays may be located in between the objective lens and the erector and/or between the erector and ocular lens, for example, depending on if a magnification of the reticle is desired. The display may further be integrated with an optical component of the sight or integrated with a transparent optical protection component. The sighting system may further include an adjustment system for adjusting the reticle or other information to be displayed on the display.

(8) Additional advantages and novel features of these aspects will be set forth in part in the description that follows, and in part will become more apparent to those skilled in the art upon examination of the following or upon learning by practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more example aspects of the present disclosure and, together with the detailed description, serve to explain their principles and implementations.

- (2) FIG. 1 is a rear perspective view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (3) FIG. 2 is a rear perspective view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (4) FIG. 3 is a rear perspective view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (5) FIG. 4 is a front perspective view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (6) FIG. 5 is a side view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (7) FIG. 6 is a side view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (8) FIG. 7 is a rear view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (9) FIG. 8A is a rear view depicting an example unmounted scope and integrated mount in accordance with one aspect of the present disclosure;
- (10) FIG. 8B is a rear view depicting an example mounted scope and mounting assembly in accordance with one aspect of the present disclosure;
- (11) FIG. 9 is a simplified cross section view depicting one example of a scope in accordance with one aspect of the present disclosure;
- (12) FIG. 10 is a simplified cross section view depicting one example of a scope in accordance with one aspect of the present disclosure;
- (13) FIG. 11A is a view of an example reticle display in accordance with one aspect of the present disclosure;
- (14) FIG. 11B is a view of an example reticle display having reference dots and magnified view thereof in accordance with one aspect of the disclosure;
- (15) FIG. 12A is a view depicting a first example controller in accordance with aspects of the present disclosure;
- (16) FIG. 12B is a view depicting a second example controller in accordance with aspects of the present disclosure;
- (17) FIG. 13 is a flow chart depicting one control method of the reticle display in accordance with one aspect of the present disclosure;
- (18) FIG. 14 is a view of an example reticle display in accordance with one aspect of the present disclosure;
- (19) FIG. 15 is a view of an example reticle display in accordance with one aspect of the present disclosure.
- (20) FIG. 16 contains a representative diagram of an example computer system capable of carrying out functionality described in example implementations via a coupled device, such as a device on a network, in accordance with aspects of the present invention.

DETAILED DESCRIPTION

(21) In connection with the views and examples of FIGS. 1-9, wherein like numbers indicate the same or corresponding elements throughout the views, FIGS. 1-7 show views of an example scope assembly **100** with integrated mount **141** and a mounting point **110** for the scope assembly **100**, in accordance with aspects of the present disclosure. The scope assembly **100** may include a main scope body **122** having an inner (FIG. 8) **121** and an outer surface **120**. The main scope body **122** is shown in FIG. 1 as having a generally tubular outer shape; however, the scope body is not limited as such shape.

(22) The main scope body **122** may comprise a composite and/or a composite substrate. A composite for use with this example implementation is not limited to, and may include any one or combination of: carbon fiber, carbon fiber reinforced plastic, poly-para-phenylene terephthalamide,

glass-reinforced plastic, and/or glass-fiber reinforced plastic. The body **122** may further include an epoxy, polyester, vinyl ester, and/or nylon. The main scope body **122** may be formed as a molded fiber reinforced plastic, layered and/or laminated substrate or a molded thermoplastic matrix, for example. Further, the inner surface of the main scope body may include a threaded and/or machined insert **128** proximal to the inner surface of the scope body, for example, for mounting of components inside the scope body. The insert **128** may comprise: aluminum, titanium, steel, brass, silicone bronze, and/or an alloy of the aforementioned materials, for example. The insert **128** and inner scope body may include additional features and/or processing to improve the interface between materials. Non-limiting examples of additional features and/or processing may include anodization and/or the addition of a separator between the outer scope body and the insert **128** (such as a fiberglass scrim), which may be added to prevent corrosion or galling between the insert and the outer scope housing. The insert **128**, alternatively may comprise polyether ether ketone, reinforced polyether ether ketone, polyetherimide, reinforced polyetherimide, and/or reinforced nylon, for example.

(23) The scope assembly **100** may further include an objective bell **136** and an eyepiece **126**. The abovementioned insert **128** may further may include at least one transparent rear optic or optic protection portion **130**. The scope assembly may further include one or more adjustment turrets or knobs; such as an elevation adjustment turret **132** and/or a windage adjustment turret **134**. The turrets **132**, **134** may control the location of the reticle for elevation and/or windage, for example. The assembly may further include a combination turret (not shown) for control of reticle illumination and/or image focus. The abovementioned turrets may also or alternatively adjust the erector **244**, **246** and/or **248** (FIG. **9**) to adjust for windage and/or elevation.

(24) Further, the abovementioned turrets **132**, **134** may have one or more detents for providing tactile feedback when a user turns the turret. The turrets **132**, **134** are not limited to knobs as shown, and may provide a similar function in the form of a slider, button, or rocker switch, for example. The scope assembly **100** may further include a magnification ring **124** for varying the magnification of the scope.

(25) The scope assembly **100** may further include a mounting portion **148** integrally mounted to the main scope body **122**. The mounting portion **148** may be integrally molded as a part of the scope assembly **100** or may, for example, be bonded or otherwise attached to the scope body **122** or another portion of the scope assembly **100**. The mounting portion may be formed of a composite that is the same as that of much of the remainder of the scope assembly **100**, or may be formed of a different material from the scope assembly. Since the mounting portion may be mounted to an aluminum rail (e.g., for use with a firearm), it may be preferable to use a material similar in electropotential or that is substantially non-conductive in order to prevent galvanic corrosion therebetween. Further it may be advantageous to form the mounting portion using a lightweight material that has a low coefficient of expansion, for example. It may be preferable to form the mounting portion of a material having a coefficient of linear thermal expansion less than 55×10^{-6} m/mk ($\pm 10\%$) between 23° C. and 150° C. The mounting portion may comprise at least one of a carbon fiber, carbon fiber reinforced plastic, poly-para-phenylene terephthalamide, glass-reinforced plastic, glass-fiber reinforced plastic, polyether ether ketone, reinforced polyether ether ketone, polyetherimide, reinforced polyetherimide, and/or reinforced nylon, for example. The composite may further include; epoxy, polyester, vinyl ester, nylon. The use of polyether ether ketone, reinforced polyether ether ketone, polyetherimide, reinforced polyetherimide, and/or reinforced nylon may prevent corrosion between the mounting portion **148** and the metallic mounting portion of the firearm, for example. The mounting portion **148** may further be either molded as a fiber reinforced plastic and/or be comprised of a layered or laminated substrate. Forming the mounting portion **148** of polyether ether ketone, reinforced polyether ether ketone, polyetherimide, reinforced polyetherimide, and/or reinforced nylon, may further prevent corrosion between the mounting portion **148** if the mounting rail **105** is formed of an aluminum or similar

metallic material, for example.

(26) The mounting rail **105**, to which an integrated mount **141** may be mounted, may include a scope attachment portion **112** as shown in FIGS. **6** and **8B** which, for explanation and orientation purposes, may extend in a direction **F**. The scope attachment portion **112**, may include a plurality of recoil grooves **116** formed such that each groove extends in a direction generally perpendicular to the direction **F**. The mounting rail **105** may further include an angled portion **113** having an angled surface relative to a plane extending along an edge of the scope attachment portion **112** in the direction **F** (e.g., the upper edge of scope attachment portion **112** as shown in FIG. **8**). The angled portion **113** may be on both sides of the rail and angled in relation to an axis perpendicular to the **F** direction. The mounting rail **105** may also include second angled portions **135** as shown in FIG. **8**. The second angled portions **135** may intersect with the angled portions **113** forming a line along the intersection of the second angled portion **135** and the angled portion **113**, the line extending substantially perpendicular to the direction **F**. The second angled portion **135** and may be on both sides of the rail in relation to an axis perpendicular to the direction **F**.

(27) Referring to FIGS. **8A** and **8B**, the integrated mount **141** may include a receiving portion **160** for receiving the mounting rail **105**. The receiving portion **160** may include a mounted flat portion **148** which may extend in direction **F** as shown in FIG. **6** when aligned with the mounting rail **105**. The receiving portion **160** may further include a first angled portion **150**. The mounted flat portion **148** may intersect with the first angled portion **150** as shown in FIG. **8A** forming edge extending in a direction parallel to direction **F** when the scope is oriented as shown in FIG. **2**. The first angled portion **150** may further be located on both sides of the flat portion **148** and angled in relation to an axis perpendicular to direction **F** when the flat portion is aligned with the mounting rail **105** and when the scope and firearm body are oriented as shown in FIG. **2**. The receiving portion **160** may further include a clamp mount **144**. The clamp mount **144** may include an inner clamp wall **146** that may intersect and may be angled with relation to the first angled portion **150**. The clamp mount **144** may further include a single or a plurality of clamping points **137** (FIG. **5**) and **140** (FIGS. **5** and **8A**). The clamp points **137**, **140** may include, but are not limited to, a threaded through hole extending from an outer surface **170** of the integrated mount **141** to the inner clamp wall **146**. Wherein the clamp point may be configured to receive a corresponding threaded clamp (not shown), that may be a bolt for example. The clamp may thread through the threaded through hole and contact the mounting rail **105** at a second angled portion **135**. The clamp may **144** further include a retainer portion to prevent the accidental removal of the clamp from the threaded portion **140**; as an example, the retainer portion may comprise a stopper on the end of a bolt, a washer or some other method of preventing the example bolt from backing all the way out of the threaded through hole **140**. Other clamping methods may be employed such as a cam or cam mechanism for tightening the receiving portion **160** to the mounting rail **105**. As shown in FIG. **8B**, the clamp mount **144** may allow for a force in a direction **200**, having components in directions **213** and **214**; force **214** causing the integrated mount to self-center on the mounting rail **105**. Self-centering may occur through the abovementioned force **214** and the interaction between the angled portions **113** of the mounting rail **105** and the first angled portions **150** of the integrated mount **141**.

(28) The integrated mount may further include a recoil stop **142**, which passes through a through hole on both ends of the integrated mount **141**, A shown in FIG. **8A**, the recoil-stop may extend between a recoil groove **116** and interoperate with and may contact a recoil groove **116** of the mounting rail **105**. The recoil stop **142** may comprise, but is not limited to, a pin or a bolt. The recoil stop **142** may further include a retainer portion **117**, or preventing the recoil stop **142** from being removed once the stop is installed; the retainer portion **117** may comprise a pin extending through a hole at portion **117** on the end of the recoil stop, for example.

(29) The scope body **122** may optionally house an electroluminescent electronically adjustable reticle **800**. As shown in FIG. **11**, the electroluminescent adjustable reticle may comprise, for example, an optically transparent substrate **805**, having a substantially transparent optical array

801. The substrate may be formed according to the method described at the website: <http://lumineq.com/en/products/tfel>; as viewed on Jul. 14, 2016, the contents of which are hereby incorporated by reference in its entirety. Referring now to FIGS. **9** and **10**, the scope may include a first end O and a second end S. The substrate **805** may be located in a focal plane, such as at a point between the objective lenses and ocular lens or lenses for example, in a location **350** between an erector **248** and an ocular lens or lenses **252** and **253**. An erector lens assembly **244**, **246** and/or **248** may be located between the objective lens or lenses **280** or ocular lenses **252** and/or **253**, and the transparent substrate **805** may be located between the objective lens and the erector in region **320** and/or between the erector and ocular lens in region **340** depending on if a magnification of the reticle is desired. The optically transparent substrate **805** may further be incorporated into any of the optics within the sight, some non-limiting examples being the objective lens **280**, ocular lens or lenses **152** and/or **252**, and/or erectors **244**, **246** and **248**, or the substrate **805** may be integrated into the transparent optics protection portions **130** and/or **161** at either end of the scope.

(30) Electroluminescent displays are generally comprised of a substrate having a layer of electroluminescent material between two conductive layers. When current is applied across the conductive layers of the substrate the electroluminescent material emits visible light. In one aspect, the electroluminescent display may comprise a glass or transparent substrate having thin film on the glass or substrate. The thin film may include a layer of light emitting electroluminescent material formed thereon or attached thereto, such light emitting electroluminescent material may comprise, for example, a phosphor; manganese, silver, or copper-doped zinc sulfide. The electroluminescent layer may be located or formed between two dielectric layers that comprise the electroluminescent display. Electric current passed through or a magnetic field imposed upon the display may cause the abovementioned electroluminescent layer to give off photons and thereby emit visible light, for example. The abovementioned conductive layers may be a transparent electrode formed of a Tin Oxide or other conductive materials, as described further below, or may be provided through metal nanofibers for example. The display may be formed according to the processes, or used in conjunction with various features, described in U.S. Pat. No. 9,226,362 B2 and U.S. Pat. No. 9,290,840 B2, which are attached and hereby incorporated by reference.

(31) Referring now to FIGS. **11**, **14** and **15** The optical array **801** may include a plurality of lines **820** extending in a first direction V (e.g., extending vertically when viewing the array **801** when positioned on a firearm in a firing position) that remain substantially transparent when not in use but one or more of which may become lit or otherwise highlighted **950** (FIG. **15**) when electric current or magnetic field is applied. The optical array **801** may further include a plurality of second lines **810** extending in a second direction H (e.g., extending horizontally when viewing the array **801** when positioned on a firearm in a firing position) that remain transparent when not in use but one or more of which may become lit or highlighted **810** (FIG. **14**) when electric current or a magnetic field is applied. A controller **700**, **750** (FIGS. **12A**, **12B**), for example, may selectively apply current or voltage, for example, to each of the horizontal or vertical lines **810**, **820** to be lit or highlighted through a display interface **716** (FIG. **12A**). For example, the controller **700**, **750** (see, e.g., FIGS. **12A**, **12B**, described further below) may selectively provide current or voltage to each display segment to be lit or highlighted based on a user operating the abovementioned turrets **132** and **134** (e.g., FIG. **1**). As shown in the process **900** of FIG. **13**, the controller may receive an input signal **910** from turrets **132** and **134** (FIG. **1**), and based on the input signal received make a first determination **920** to adjust the location of the current or voltage to be applied to a particular segment **820** (FIG. **15**) in a horizontal direction **950** (FIG. **15**). For example, as shown in FIG. **15**, the current or voltage may be initially applied as a default to a vertically extending segment in a first position **820**, and move to a second position **820a** or a third position **820b** in response to windage adjustment received from the windage turret **134** (FIG. **1**). Further the controller may receive an input signal **910** from the abovementioned turrets **132** and **134** (e.g., the elevation turret **132** of FIG. **1**), and based on the received signal make a second determination **930** to adjust the

location of the current or voltage applied to from an initial default segment **810** (FIG. **14**) in a vertical direction **940** (FIG. **14**). For example, as shown in FIG. **14**, the current or voltage may be applied so as to change the lit or highlighted segment from a first position **810** to a second position **810a** or a third position **810b**. The controller **700**, **750** (FIG. **12A** or **12B**) may adjust the abovementioned location of the currents based on a manual input from the user (e.g., the abovementioned turrets **132** and/or **134**) or automatically based on inputs regarding any one of a windage, elevation, and/or distance determination, for example. Further, the controller **700**, **750** (FIG. **12A** or **12B**) may provide an auto-off feature for turning off illumination of any sections of the optical array when the scope is stationary for a set period of time. The controller **700**, **750** (FIG. **12A** or **12B**) may further feature an auto-on/off feature based on the detection of movement of the scope or detection of a user placing the eyepiece **126** (FIG. **1**) in proximity to the user's eye, for example. Such detection may be made, for example, using an inertial or other motion sensor for movement, or an infrared or other proximity detector for determining proximity to a user's eye as described in U.S. Pat. No. 5,699,115 which is attached and hereby incorporated by reference, for example.

(32) The optical array **801** may optionally further include any number of informative displays **830** for displaying characters, for example, as shown in FIG. **11**. The informative display may display, but is not limited to numerals relating to a yardage or other distance, round count, or elevation measurement. The informative display **830** may be formed using the methods described above (and devices described below) and/or via use of a similar thin film substrate as is used for the abovementioned line portions. The informative display **830** may further include a plurality of the abovementioned informative displays. The illumination of the informative display may be controlled by the abovementioned controller **700**, **750** (see FIGS. **12A** and **12B** and further accompanying description below), or via a separate controller or other device.

(33) In another aspect, the adjustable reticle may comprise, for example, a series of reference dots on each of the abovementioned segments to allow a user to view a frame of reference with or without adjusting the reticle. For example, as shown in FIGS. **10A-C**, a series of first reference dots **860** may be part of and/or added and arranged in the first direction V along at least one of the plurality of lines **820** (e.g., extending vertically when viewing the array **801** when positioned on a firearm in a firing position). The reticle may further include a series of second reference dots **870** which may be spaced at intervals along at least one of the plurality of second lines **810** in a second direction H (e.g., extending horizontally when viewing the array **801** when positioned on a firearm in a firing position). The reference dots **860** and/or **870** may remain substantially transparent when not in use, but one or more of which may become lit or otherwise highlighted when the line segment along which the reference dots are arranged is lit. Further, the reference dots may be capable of being individually lit either with or separate from the line segment along which the reference dots are arranged. In one aspect each individual reference dot may be lit individually or by selected groups based on input from the controller **700**.

(34) As shown in FIGS. **10B** and **10C**, the reference dots may be evenly spaced, for example, such that there are one or a plurality (e.g., four) dots on a segment in the first direction V between two respective segments in the second direction H and vice versa. The MIL-Dots may be spaced, for example, so as to allow for range estimation and/or to compensate for bullet drop and/or windage adjustments. As an example, the reference dots may be MIL-Dots and the spacing between each MIL-Dot may represent a miliradian. As an example, the MIL-Dots may be spaced so that the spacing between the centers of two consecutive MIL-Dots represents 1 yard at a distance of 1000 yards. It is noted, that the reference dots may be located along any of the above segments and/or may be located at any separate location on the reticle. Further, in one example implementation, the reference dots may be etched into the surface of the reticle substrate or printed on the reticle surface so as to be visible at all times through the scope.

(35) As shown in FIG. **12A**, in one example implementation, the controller **700** may include a

processor **712**, a power supply **714**, a display interface **716** for controlling a display, such as an electroluminescent electronically adjustable reticle **800**, and a memory **740**. Example features of each of these portions of the controller **700** of FIG. **12A** is discussed in more detail with regard to FIG. **12B** below.

(36) As shown in FIG. **12B**, the controller **750** may alternatively include or be coupled with a computer, or aspects thereof (e.g., via wired or wireless coupling, such as is shown in and described with respect to FIG. **16** below), for implementing the abovementioned steps using hardware, software or a combination thereof. In one aspect of the present disclosure, features are directed toward one or more computer systems capable of carrying out the functionality described herein.

(37) Computer system **600** includes one or more processors, such as processor **604**. The processor **604** may be connected to a communication infrastructure **606** (e.g., a communications bus, cross-over bar, or network). Various software aspects are described in terms of this exemplary computer system. After reading this description, it will become apparent to a person skilled in the relevant art(s) how to implement aspects of the invention using other computer systems and/or architectures.

(38) Computer system **600** can include a display interface **602** that translates data from the communication infrastructure **606** (or from a frame buffer not shown) for display on the reticle **800**. Computer system **600** also includes a main memory **608**, which may be a random access memory (RAM), and may also include a secondary memory **610**. The secondary memory **610** may include, for example, a hard disk drive **612** and/or a removable storage drive **614**, representing a universal serial bus (USB) flash drive, etc. The removable storage drive **614** reads from and/or writes to a removable storage unit **618** in a well-known manner. Removable storage unit **618** may include a USB flash drive, for example, which is read by and written to removable storage drive **614**. As will be appreciated, the removable storage unit **618** includes a computer usable storage medium having stored therein computer software and/or data.

(39) Alternative aspects of the abovementioned computer may include secondary memory **610** and may include other similar devices for allowing computer programs or other instructions to be loaded into computer system **600**. Such devices may include, for example, a removable storage unit **622** and an interface **620**. Examples of such may include a program cartridge and cartridge interface, a removable memory chip (such as an erasable programmable read only memory (EPROM), or programmable read only memory (PROM)) and associated socket, and other removable storage units **622** and interfaces **620**, which allow software and data to be transferred from the removable storage unit **622** to computer system **600**.

(40) The computer **600** in controller **700** may also include a communications interface **624**. Communications interface **624** allows software and data to be transferred between computer system **600** and external devices. Examples of communications interface **624** may include a Bluetooth® module, a network interface, a communications port, a Personal Computer Memory Card International Association (PCMCIA) slot and card, etc. Software and data transferred via communications interface **624** are in the form of signals **628**, which may be electronic, electromagnetic, optical or other signals capable of being received by communications interface **624**. These signals **628** are provided to communications interface **624** via a communications path (e.g., channel) **626**. This path **626** carries signals **628** and may be implemented using wire or cable, fiber optics, a telephone line, a cellular link, a radio frequency (RF) link and/or other communications channels. In this document, the terms “computer program medium” and “computer usable medium” are used to refer generally to media such as a removable storage drive, a hard disk installed in hard disk drive, and signals **628**. These computer program products provide software to the computer system **600**. The disclosure is directed to such computer program products.

(41) Computer programs (also referred to as computer control logic) are stored in main memory **608** and/or secondary memory **610**. Computer programs may also be received via communications interface **624**. Such computer programs, when executed, enable the computer system **600** to

perform the features in accordance with aspects of the present disclosure, as discussed herein. In particular, the computer programs, when executed, enable the processor **604** to perform the features in accordance with aspects of the present disclosure. Accordingly, such computer programs represent controllers of the computer system **600**.

(42) In an aspect of the present disclosure where the steps are implemented using software, the software may be stored in a computer program product and loaded into computer system **600** using removable storage drive **614**, hard drive **612**, or communications interface **620**. The control logic (software), when executed by the processor **604**, causes the processor **604** to perform the functions described herein. In another aspect of the present disclosure, the system is implemented primarily in hardware using, for example, hardware components, such as application specific integrated circuits (ASICs). Implementation of the hardware state machine so as to perform the functions described herein will be apparent to persons skilled in the relevant art(s). In another aspect of the present disclosure, the disclosed controller functions may be implemented using a combination of both hardware and software.

(43) The substrate **805** of reticle **800** may further be used in combination with a wire reticle, a cross-hair, and/or a reticle etched into the substrate that is/are provided as a stationary frame of reference in case of an electroluminescent display failure or misalignment.

(44) FIG. **16** shows an example components within a system **1100** usable in accordance with the present invention. The system **1100** includes one or more users **1160**, **1162**, one or more controller components within a scope **1142** and one or more terminals **1166**. In one aspect, data input for use, and various features for processing and display, for example, in accordance with aspects of the present invention may be, for example, input and/or accessed by the one or more controller components within the scope **1142**, which may further be processed or otherwise communicate with one or more terminals **1166**, such as personal computers (PCs), minicomputers, mainframe computers, microcomputers, telephonic devices, or wireless devices, such as personal digital assistants (“PDAs”), smart phones, or other hand-held wireless devices, which may optionally be coupled to a server **1143**, such as a PC, minicomputer, mainframe computer, microcomputer, or other device having a processor and a repository for data and/or connection to a repository for data, via, for example, a network **1144**, such as the Internet or an intranet, and couplings **1145**, **1146**, **1164**. The couplings **1145**, **1146**, **1164** include, for example, wired, wireless, or fiberoptic links.

(45) While the aspects described herein have been described in conjunction with the example aspects outlined above, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that are or may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example aspects, as set forth above, are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later-developed alternatives, modifications, variations, improvements, and/or substantial equivalents.

Claims

1. An optical sighting apparatus and mount, comprising: a main body, with a first end portion and a second end portion and having a central axis extending from the first end portion to the second end portion; an objective lens disposed within the body; an eyepiece housing a lens, the mount is formed of a substantially non-conductive composite material and configured to be permanently bonded to the main body, the mount comprising: a rail receiver portion for receiving a mounting rail, wherein the rail receiver portion further comprises: at least one flat portion forming a plane having a major axis perpendicular to the central axis; a first angled portion located on each end of a minor axis of the flat portion, the first angled portion being capable of receiving a respective angled portion of a receiver portion of the mounting rail; at least one second angled portion intersecting

with at least one of the first angled portions; and a clamping portion with a threaded mechanism to clamp the rail receiver portion to the mounting rail, wherein the threaded mechanism of the clamping portion comprises a bolt with a contact portion configured to contact the mounting rail to clamp the optical sighting apparatus to the mounting rail, and rotation of the bolt to clamp the optical sighting apparatus to the mounting rail self-centers the optical sighting apparatus with respect to the mounting rail.

2. The optical sighting apparatus and mount of claim 1, wherein a first force and a second force supplied by the first angled portion center the optical sighting apparatus with respect to the mounting rail when the bolt is rotated.

3. The optical sighting apparatus and mount of claim 1, wherein the clamping portion comprises a threaded hole for receiving the threaded bolt; and a retainer portion for preventing the threaded bolt from separating from the threaded hole.

4. The optical sighting apparatus and mount of claim 1, wherein the mount is formed of at least one of a carbon fiber, carbon fiber reinforced plastic, poly-para-phenylene terephthalamide, glass-reinforced plastic, glass-fiber reinforced plastic, polyether ether ketone, reinforced polyether ether ketone, polyetherimide, reinforced polyetherimide, or reinforced nylon.

5. The optical sighting apparatus of claim 1, wherein the mount is formed of a material having a coefficient of linear thermal expansion less than 55×10^{-6} m/mk between 23° C. and 150° C.

6. The optical sighting apparatus and mount of claim 1, wherein the mounting rail is attached to a firearm.

7. A mount configured to be permanently bonded to an optical sighting apparatus, the mount comprising: a mount body having a first end portion and a second end portion with a central axis extending therebetween and formed of a first composite material and configured to be permanently bonded to the optical sighting apparatus; a rail receiver portion for receiving a mounting rail, wherein the rail receiver portion further comprises: at least one flat portion forming a plane having a major axis perpendicular to the central axis; a first angled portion located on each end of a minor axis of the flat portion, the first angled portion being capable of receiving a respective angled portion of the mounting rail; at least one second angled portion intersecting with at least one of the first angled portions; and a clamping portion with a threaded mechanism to clamp the rail receiver portion to the mounting rail, wherein rotation of the threaded mechanism provides a clamping force to the mounting rail and self-centers the mount on the mounting rail.

8. The mount of claim 7, wherein the mount is formed of at least one of a carbon fiber, carbon fiber reinforced plastic, poly-para-phenylene terephthalamide, glass-reinforced plastic, glass-fiber reinforced plastic, polyether ether ketone, reinforced polyether ether ketone, polyetherimide, reinforced polyetherimide, or reinforced nylon.

9. The mount of claim 7, wherein the mount is formed of a material having a coefficient of linear thermal expansion less than 55×10^{-6} m/mk between 23° C. and 150° C.

10. The mount of claim 7, wherein the mounting rail is attached to a firearm.

11. The mount of claim 7, further comprising an optical sighting apparatus permanently connected thereto, wherein the optical sighting apparatus comprises: a main body, with a first end portion and a second end portion and having a central axis extending from the first end portion to the second end portion; an objective lens disposed within the body; and an eyepiece housing a lens.

12. The mount of claim 11, further comprising an erector lens disposed within the body at a position between the objective lens and the eyepiece lens.

13. The mount of claim 12, wherein the optical sighting apparatus is formed of a second composite material is different from the first composite material of the mount.

14. The mount of claim 7, wherein the first composite material is non-conductive.

15. The mount of claim 7, wherein the clamping portion comprises a first threaded through hole for receiving the threaded mechanism, wherein the axis of the threaded hole is angled with respect to the first flat portion so that a portion of the threaded mechanism contacts a second angled portion

of the mounting rail.

16. A mount configured to be permanently bonded to an optical sighting apparatus, the mount comprising: a mount body having a first end portion and a second end portion with a central axis extending therebetween and formed of a first composite material and configured to be permanently bonded to the optical sighting apparatus; a rail receiver portion for receiving a mounting rail, wherein the rail receiver portion further comprises: at least one flat portion; a first angled portion located on each end of a minor axis of the flat portion, the first angled portion being capable of receiving a respective angled portion of a receiver portion of the mounting rail; at least one second angled portion intersecting with at least one of the first angled portions; and a clamping portion with a threaded mechanism to clamp the rail receiver portion to the mounting rail, wherein rotation of the threaded mechanism clamps the mount to the mounting rail, wherein the threaded mechanism of the clamping portion is a bolt configured to contact the mounting rail.

17. The mount of claim 16, further comprising a second threaded mechanism, wherein the second threaded mechanism comprises a second bolt to clamp the optical sighting apparatus to the mounting rail.

18. The mount of claim 17, wherein the clamping portion comprises a first threaded hole for receiving the threaded bolt and a second threaded hole for threadably receiving the second bolt, wherein the second bolt is configured to contact the mounting rail.

19. The mount of claim 16, wherein the clamping portion comprises a threaded hole for receiving the threaded bolt; and a retainer portion for preventing the threaded bolt from separating from the threaded hole.

20. The mount of claim 16, wherein the clamping portion comprises a first threaded through hole for receiving a threaded bolt, wherein the axis of the threaded hole is angled with respect to the first flat portion so that a portion of the bolt contacts a second angled portion of the mounting rail.
